

AI Failures Museum

COM2020 Coursework 2: v1.0.0

Java_The_Hutt

March 23, 2026

The Problem

- AI systems increasingly support environmental decision-making
- Real deployments often fail due to incomplete data, bias, or misuse
- These failures are rarely documented in accessible ways

Our Solution

- Web-based museum of AI failure case studies
- Interactive exhibits teaching how and why systems fail
- Support for visitors (learners), curators (moderators), and developers
- Focus on learning and responsibility, not blame

Chase Sibley - Scrum Master & Testing Lead

Responsibilities

- Sprint planning and meeting minutes documentation
- Test suite design and implementation
- Quality assurance and test coverage

Key Deliverables

- Automated tests: 90 tests, covering web-health, object creation, feature management & RBAC
- Manual test plan: End-to-end coverage
- CI/CD pipeline: Passing

GitHub Evidence

- Commits: 4b564f0 (web-server), 4af61cb (comments), ac9d0ec (authentication), 4badf3d (accounts), 418b0a5 (RBAC), abeb4f2 (quizzes) & 4388f79 (bookmarking)
- Framework: pytest

Oliver Methold - Product Owner

Responsibilities

- Role assignment and team coordination across all sprints
- Progress monitoring and quality oversight of team deliverables
- Stakeholder communication and final decision-making authority

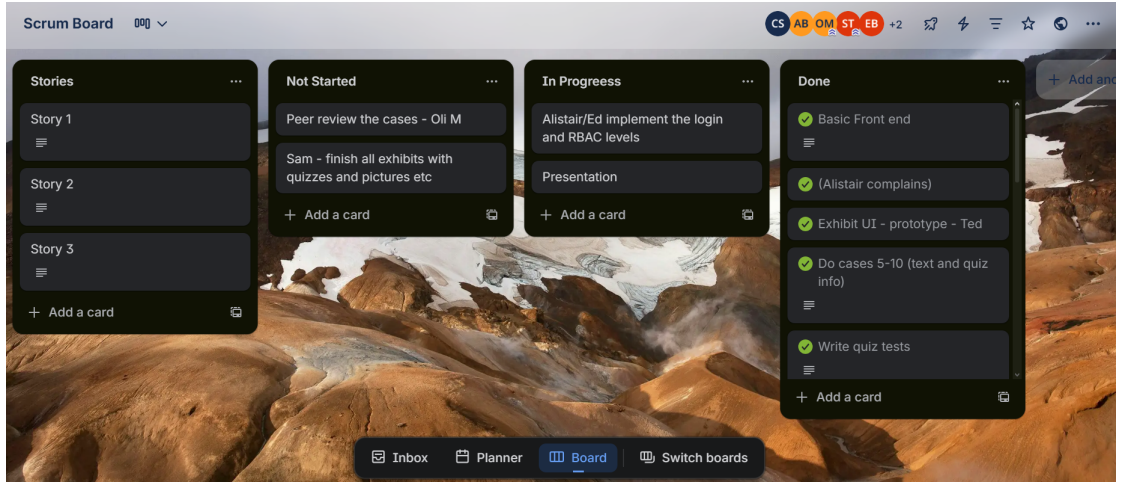
Key Deliverables

- Requirements prioritised and communicated to the team
- Kanban board maintained with clear task ownership and deadlines
- Ensured smooth inter-team communication throughout both sprints

GitHub Evidence

- Issues created and managed throughout the project lifecycle
- Kanban board: Active and maintained across Sprint 1 and Sprint 2 (Trello)

Scrumboard (Trello)



LIVE DEMO

Technical Architecture & System Walkthrough

(Alistair Bishop – Technical Lead)

Browse exhibits → View case study → Complete quiz → Curator tools

Responsibilities

- User interface design and prototyping
- Accessibility compliance
- User experience and usability testing

Key Deliverables

- Design files: Wireframes
- Accessibility: WCAG Level A

UI Design Choices

Design Goals

- Clear, modern, approachable interface
- Break complex content into smaller sections
- Keep focus on content, not clutter

Key Choices

- Strong visual hierarchy (headings + cards)
- Consistent layout across all pages
- Search, filters, and tags for quick scanning
- Subtle colours and icons for engagement

Improving User Experience

Easier to Digest

- Structured sections (context, failure, impact)
- Card layout + spacing for readability
- Quizzes and bookmarks reinforce learning

Usability

- Simple, consistent navigation
- Accessible and readable structure
- Personal features (scores, saved items)

Sam Taylor-King - Data & Analytics Lead

Responsibilities

- Case study creation and curation (realistic & ethical)
- Role-Based Access Control (RBAC) system design
- User feedback collection and product evaluation

Key Deliverables

- **20 exhibits:** 11 demo cases + 9 production cases (5 real-world with citations)
- **Interactive quizzes:** One per exhibit for engagement
- **RBAC system:** User, Curator, Admin roles enabling live website editing
- **User evaluation:** Google Form feedback + UI improvement proposal

GitHub Evidence

- Commits: 8ea3157 (cases), d5d36a6 (RBAC & real cases), 52453d9 (ticket), b569bae (splash image)
- Outcome: Shifted from backend-only to live curator editing; improved UI engagement

Responsibilities

- UI component implementation: Django templates, custom css stylesheet, inline styling
- API integration with backend: Django URL routing
- State management and frontend architecture: Server side rendering Request/response

Key Deliverables

- Components built: 14 html templates for UI
- API integration: paths for each page in urls.py
- Responsive design: views.py to process requests and load data

Responsibilities

- Final report production (evaluation, design trade-offs, ethics update, future roadmap)
- Technical documentation coordination across team
- Ethical and legal considerations documentation

Key Deliverables

- **Sprint 1:** Prototype report, data model documentation, ethics document, risk register
- **Sprint 2:** Final report with usability evaluation

GitHub Evidence

- Commits: 3c3fe2f, documentation structure across both sprints
- Verified database facts: 3 quiz attempts (77.8% avg), 2 users

Key Achievements & Evaluation

From CW1 to CW2

- Expanded exhibit dataset from prototype to 20 fully structured case studies
- Curator live editing enabled via RBAC system replacing admin-only workflow
- Quiz, comment, and bookmarking features fully integrated end-to-end

Evaluation Results

- 90 automated tests passing across authentication, RBAC, quizzes, and comments
- User feedback collected via Google Form; informed UI improvements
- 3 user roles (visitor, curator, admin) successfully implemented and verified

Deployment Status

- Local run guide: Complete and documented in README
- Seeded dataset: 20 exhibits, 60+ quiz questions included out of the box

Questions?

Thank you